



## BENTOML CHEAT SHEET

Core setup, initialization, operations &amp; integration quick reference

PART 1: SETUP &amp; CORE

## 01. OVERVIEW &amp; SETUP

Installation

BentoML AI/ML framework

```
pip install bentoml
# Verify installation
python -c "import bentoml; print('OK')"
# GPU support check
import torch; print(torch.cuda.is_available())...
```

Use virtual environment: `python -m venv venv`

## 04. TRAINING LOOP

Workflow

```
optimizer = torch.optim.Adam(model.parameters...)
criterion = nn.CrossEntropyLoss()
for epoch in range(num_epochs):
    for batch_X, batch_y in dataloader:
        optimizer.zero_grad()
        output = model(batch_X)
        loss = criterion(output, batch_y)
        loss.backward()
        optimizer.step()
```

## 02. DATA PREPARATION

Architecture

```
import pandas as pd
df = pd.read_csv('data.csv')
X = df.drop('target', axis=1).values
y = df['target'].values
from sklearn.model_selection import train_test...
X_train, X_test, y_train, y_test = train_test...
```

Normalize/standardize features for best results

## 05. EVALUATION

CRUD API

```
from sklearn.metrics import accuracy_score, c...
model.eval()
with torch.no_grad():
    preds = model(X_test)
    accuracy = accuracy_score(y_test, preds.arga...
    print(classification_report(y_test, preds.arg...
```

Track precision, recall, F1, use confusion matrix

## 03. MODEL DEFINITION

YAML / Config

```
# Simple neural network:
import torch.nn as nn
class Model(nn.Module):
    def __init__(self):
        super().__init__()
        self.layers = nn.Sequential(
            nn.Linear(input_size, 128),
            nn.ReLU(),
            nn.Linear(128, num_classes)
        )
    def forward(self, x): return self.layers(x)
```

## 06. INFERENCE &amp; SERVING

Integration

```
# Save model
torch.save(model.state_dict(), 'model.pt')
# Load model
model.load_state_dict(torch.load('model.pt'))
model.eval()
# Inference
with torch.no_grad():
    prediction = model(input_tensor)
```



# SECURITY, MONITORING & OPERATIONS

Production hardening, observability, performance tuning & CLI reference

PART 2: OPS & SECURITY

## 07. HYPERPARAMETERS

Hardening

Learning rate: start with 1e-3, tune with scheduler  
Batch size: 32-256 depending on GPU memory  
Epochs: use early stopping to prevent overfitting

```
scheduler = torch.optim.lr_scheduler.ReduceLR...  
scheduler.step(val_loss)
```

Grid Search or Optuna for systematic tuning

## 10. FRAMEWORKS & TOOLS

Unit Tests

|                    |                             |
|--------------------|-----------------------------|
| TensorFlow / Keras | production ML               |
| PyTorch            | research & NLP              |
| Scikit-learn       | classical ML                |
| Hugging Face       | pretrained transformers     |
| MLflow / W&B       | experiment tracking         |
| ONNX               | model export/portability... |

## 08. DATASETS & LOADERS

Observability

```
from torch.utils.data import Dataset, DataLoa...  
class MyDataset(Dataset):  
    def __len__(self): return len(self.data)  
    def __getitem__(self, idx): return self.data[...]  
loader = DataLoader(dataset, batch_size=32, s...
```

Pin memory for faster GPU data transfer

## 11. DEPLOYMENT

Commands

```
# Export to ONNX  
torch.onnx.export(model, dummy_input, 'model...'  
# FastAPI serving example  
@app.post('/predict')  
def predict(data: InputData):  
    tensor = preprocess(data)  
    return model(tensor).tolist()
```

Use TorchServe, BentoML, or Triton for production

## 09. OPTIMIZATION TECHNIQUES

Optimization

Mixed precision: torch.cuda.amp.autocast()",  
Gradient clipping: clip\_grad\_norm\_(model.parameters(), 1.0)",  
Batch normalization: nn.BatchNorm1d()",  
Dropout: nn.Dropout(0.5) for regularization",

```
# Learning rate warmup  
from transformers import get_linear_schedule...
```

## 12. COMMON GOTCHAS

Warning

Gradient explosion: use gradient clipping",  
Memory leak: always zero\_grad() before backward()",  
Train/eval mode: model.train() vs model.eval()",  
GPU-CPU mismatch: ensure all tensors on same device",  
Data leakage: split before normalization, not after",